Answers and talk notes

Solve and Graph Inequalities

What your student is learning

Tonight we SOLVE an inequality first, then graph its solution set — and the shading follows the SOLVED symbol, not the original.

Answers

Ask your student to explain before you look.

1. **C.** x > 6. x > 6, because 2 + 4 = 6.
2. **A.** x ≤ 6. x ≤ 6, because 30 ÷ 5 = 6.
3. **D.** Closed circle at 3, shade right. Closed circle at 3, shade right, because n ≥ 3 includes 3 and goes up.
4. **B.** k > 10. k > 10, because 5 × 2 = 10.

Talk about it

Which number goes at the circle, the one you started with or the solved one? Why?

Home mission

Look at a phone bill or gift card. Pretend a plan allows at most 20 GB and 12 are used. Write x + 12 ≤ 20 and solve it. Test one number that works and one that does not.

When they are stuck, ask

- Solve it first. What does the symbol say NOW?
- Does that symbol point toward bigger or smaller numbers?
- Test a shaded value in the original. Is it true?

Try not to say

- Shade the direction the original problem pointed.
- Inequalities are solved differently from equations.
- Once it is solved, the graph does not need checking.